

3. Пусть $n \geq 2$ — целое число, а α — произвольное действительное число. Доказать, что отображение $z^n + ne^{i\alpha}z$ конформно в круге $\{z: |z| < 1\}$.

1) $f(z)$ — регулярна в $\mathbb{D}$

$$f'(z) = nz^{n-1} + ne^{i\alpha} = n(z^{n-1} + e^{i\alpha}) \neq 0 \quad (|z| < 1, n-1 \geq 1, |e^{i\alpha}| = 1)$$

Тогда f конформно в каждой т. $\mathbb{D}$

2) Однозначность: пусть $f(z_1) = f(z_2)$, $z_1 \neq z_2$

$$z_1^n - z_2^n = ne^{i\alpha}(z_2 - z_1) \Rightarrow -ne^{i\alpha} = z_1^{n-1} + z_1^{n-2}z_2 + \dots$$

$$|-ne^{i\alpha}| = n \Rightarrow |z_1^{n-1} + z_1^{n-2}z_2 + \dots| \leq |z_1|^{n-1} + \dots < n \quad \text{н}$$

4. Доказать, что отображение $z^2 + az$ конформно в полуплоскости $\{z: \operatorname{Im} z > 0\}$ в том и только в том случае, когда выполняется неравенство $\operatorname{Im} a \geq 0$.

1) Конформность:

$\Rightarrow$

$f(z)$ — регулярна в $\mathbb{D}$

$$f'(z) = 2z + a \neq 0 \Leftrightarrow \begin{cases} 2\operatorname{Im} z + \operatorname{Im} a \neq 0 \\ 2\operatorname{Re} z + \operatorname{Re} a \neq 0 \end{cases} \Rightarrow \operatorname{Im} a \geq 0$$

2) Однозначность в $\mathbb{D}$: пусть $f(z_1) = f(z_2)$

$$z_1^2 - z_2^2 = a(z_2 - z_1) \Rightarrow z_1 + z_2 = -a, \text{ при этом}$$

$$\operatorname{Im} z_1 + \operatorname{Im} z_2 + \operatorname{Im} a > 0 \Rightarrow \text{н}$$

$\Leftarrow$ 1) $f(z)$ — рег в $\mathbb{D}$, $f'(z) = 2z + a \neq 0$, т.к. $2\operatorname{Im} z + \operatorname{Im} a \neq 0$

$$2) z_1^2 - z_2^2 = a(z_2 - z_1) \Rightarrow z_1 + z_2 + a = 0, \operatorname{Im} > 0 \Rightarrow \text{н}$$

10. Доказать, что для конформности квадратного трехчлена $az^2 + bz + c$ в выпуклой области D необходимо и достаточно, чтобы этот трехчлен был конформным в каждой точке области D .

Указание. Воспользоваться тем, что середина отрезка, соединяющего любые две точки области D , также лежит в области D .

1) $\Rightarrow$ перевернуто

2) $\Leftarrow$ однолистность: $a(z_1^2 - z_2^2) + b(z_1 - z_2) = 0 \Rightarrow$

$$\Rightarrow a(z_1 + z_2) + b = 0$$

$$f\left(\frac{z_1 + z_2}{2}\right) = a \frac{(z_1 + z_2)^2}{2} + b \left(\frac{z_1 + z_2}{2}\right) + c$$

Из вып. $f'(z) \neq 0 \Rightarrow a(z_1 + z_2) + b \neq 0$ $\checkmark$

3. Найти образ круга $\{z: |z - 1| < 2\}$ при следующих отображениях:

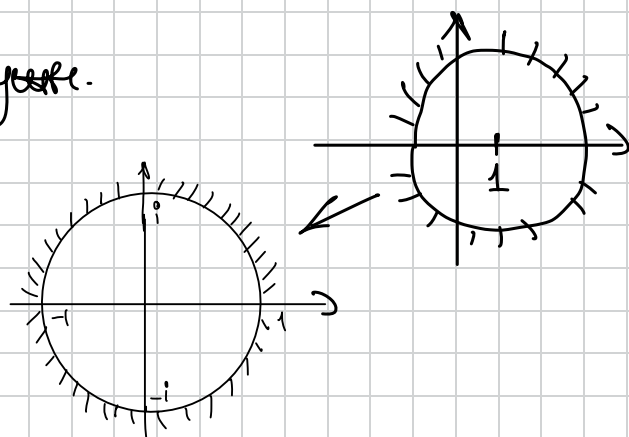
1) $w = 1 - 2iz$; ② $w = \frac{2iz}{z+3}$; 3) $w = \frac{z+1}{z-2}$; 4) $w = \frac{z-1}{2z-6}$.

$w(-3) = \infty$, $-3 \notin \Gamma \Rightarrow w(\Gamma)$ — окружность.

$|-3-1|/|a-1|=4 \Rightarrow a=0 \Rightarrow z^*=0$

$w(-1) = -i \Rightarrow$ радиус окр 1

$w(0) = 0 \Rightarrow$ сн. полюс.



6. Отыскать дробно-линейные функции $w(z)$, удовлетворяющие условиям:

1) $w(0) = 4$, $w(1+i) = 2+2i$, $w(2i) = 0$;

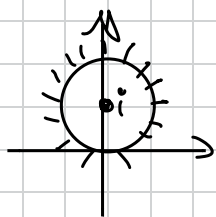
② $w(0) = 0$, $w(1+i) = 2+2i$, $w(2i) = 4$;

3) $w(0) = 0$, $w(1+i) = \infty$, $w(2i) = 2i$.

Найти образ круга $\{z: |z - i| < 1\}$ при отображениях, задаваемых этими функциями.

$$\frac{w-w_1}{w-w_2} \cdot \frac{w_3-w_2}{w_3-w_1} = \frac{z-z_1}{z-z_2} \cdot \frac{z_3-z_2}{z_3-z_1} \Rightarrow$$

$$\Rightarrow \frac{w}{w-4} \cdot \frac{-2+2i}{2+2i} = \frac{z}{z-2i} \cdot \frac{1-i}{1+i} \Rightarrow \frac{w}{w-4} = \frac{z}{z-2i} \Rightarrow w = \frac{2z}{z-i}$$

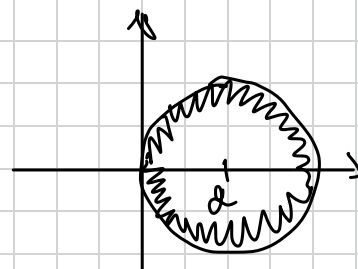


$w(i) = \infty \Rightarrow w(\text{окр})$ — прямая, $w(2i) = 4 \Rightarrow$

$\Rightarrow w(\text{окр}) = \text{OX} \Rightarrow z^* \in \mathbb{R}$

$i \notin \Gamma \Rightarrow w(\Gamma)$ — окружность

$z^* = \frac{(4-0)}{2} = 2 \Rightarrow R = |4-2| = 2$



$w(i) = \infty \Rightarrow$ подобрать внеш. точку

7. Отыскать дробно-линейные функции $w(z)$, удовлетворяющие условиям:

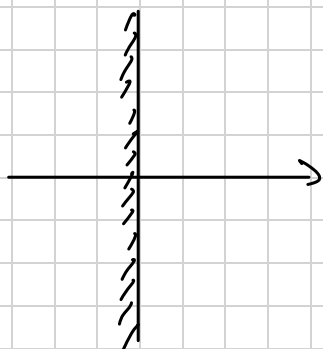
1) $w(i) = 2, w(\infty) = 1 + i, w(-i) = 0;$

2) $w(i) = 0, w(\infty) = 1, w(-i) = \infty;$

3) $w(i) = -2, w(\infty) = 2i, w(-i) = 2.$

Найти образ полуплоскости $\{z: \operatorname{Re} z > 0\}$ при отображениях, задаваемых этими функциями.

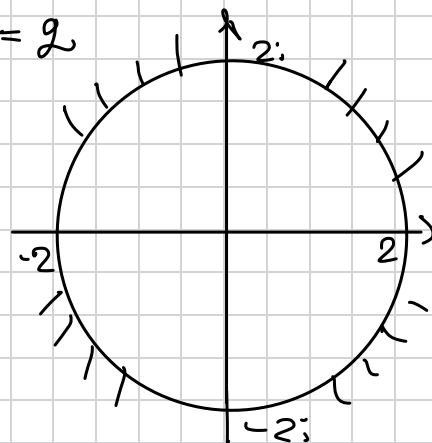
$$w = \frac{az+b}{z+d} \Rightarrow \begin{cases} \frac{a+i+b}{i+d} = -2 \\ a = 2i \\ \frac{-ai+b}{-i+d} = 2 \end{cases} \Rightarrow \begin{cases} a = 2i \\ b = -2i \\ d = 1 \end{cases} \Rightarrow w = \frac{2i z - 2i}{z+1}$$



$w(-i) = \infty, -i \notin \Gamma \Rightarrow w(\Gamma)$ — окружность

$w(0) = -2, w(i) = -2, w(-i) = 2$

$w(1) = 0 \Rightarrow$ экв. круга.



8. Найти функцию $w(z)$, конформно отображающую область D на область D_1 и удовлетворяющую указанным условиям:

1) $D = \{z: |z| < 1\}, D_1 = \{w: |w| < 1\}, w(z_0) = 0, \arg w'(z_0) = \alpha$ ($|z_0| < 1$);

2) $D = \{z: |z| < 1\}, D_1 = \{w: |w| < 1\}, w(z_0) = w_0, \arg w'(z_0) = \alpha$ ($|z_0| < 1, |w_0| < 1$);

3) $D = \{z: \operatorname{Im} z > 0\}, D_1 = \{w: |w| < 1\}, w(z_0) = 0, \arg w'(z_0) = \alpha$ ($\operatorname{Im} z_0 > 0$);

4) $D = \{z: \operatorname{Im} z > 0\}, D_1 = \{w: |w| < 1\}, w(z_0) = w_0, \arg w'(z_0) = \alpha$ ($\operatorname{Im} z_0 > 0, |w_0| < 1$);

$$2) f(z) = \frac{z-z_0}{1-\bar{z}z_0} e^{i\beta} = \frac{w-w_0}{1-\bar{w}w_0} e^{i\beta} = g(w)$$

$$f'_z(z) = g'_w(w) w'_z(z) \Rightarrow w'_z(z) = \frac{f'_z(z)}{g'_w(w)}, \quad f'_z(z_0) = \frac{e^{i\beta}}{1-|z_0|^2}$$

$$w'_z(z_0) = \frac{1-|w_0|^2}{1-|z_0|^2} e^{i(\beta-\gamma)} = \arg w'_z(z_0) = \beta-\gamma=\alpha \quad g'_w(w_0) = \frac{e^{i\gamma}}{1-|w_0|^2}$$

$$(z - z_0)(1 - w\bar{w}_0)e^{i\alpha} = (w - w_0)(1 - z\bar{z}_0)$$

$$w(1 - z\bar{z}_0 + (z - z_0)\bar{w}_0 e^{i\alpha}) = w_0(1 - z\bar{z}_0) + (z - z_0)e^{i\alpha} \Rightarrow$$

$$\Rightarrow w = \frac{(e^{i\alpha} - w_0\bar{z}_0)z + w_0 - z_0 e^{i\alpha}}{(\bar{w}_0 e^{i\alpha} - \bar{z}_0)z + 1 - z_0\bar{w}_0 e^{i\alpha}}$$

$$4) f(z) = \frac{z - z_0}{z - \bar{z}_0} e^{i\beta} = \frac{w - w_0}{1 - w\bar{w}_0} e^{i\gamma}$$

$$\frac{(z - \bar{z}_0) - (z - z_0)}{(z - \bar{z}_0)^2} e^{i\beta} = \frac{w'(1 - w\bar{w}_0) + \bar{w}_0 w'(w - w_0)}{(1 - w\bar{w}_0)^2} e^{i\gamma}$$

$$\frac{1}{z - \bar{z}_0} e^{i\beta} = \frac{w'(z_0)}{1 - w_0^2} e^{i\gamma} \Rightarrow w'(z_0) = \frac{1 - w_0^2}{2i \operatorname{Im} z_0} e^{i(\beta - \gamma - \frac{\pi}{2})} \Rightarrow$$

$$\Rightarrow \beta - \gamma = \alpha + \frac{\pi}{2} \Rightarrow \frac{z - z_0}{z - \bar{z}_0} \cdot e^{i(\alpha + \pi/2)} = \frac{w - w_0}{1 - w\bar{w}_0} \Rightarrow$$

$$\Rightarrow w = \frac{(w_0 + i e^{i\alpha})z - \bar{z}_0 w_0 - i z_0 e^{i\alpha}}{(1 + i \bar{w}_0 e^{i\alpha})z - \bar{z}_0 - i \bar{w}_0 z_0 e^{i\alpha}}$$

9. Найти общий вид конформного отображения следующих областей на кольцо $\{w: 1 < |w| < R\}$:

1) $\{z: |z - 3| > 9, |z - 8| < 16\}$;

2) $\{z: |z - 5| > 4, \operatorname{Re} z > 0\}$.

$$w = \frac{az + b}{cz + d}$$

$$z_1 = -\frac{d}{c}$$

$$z_1^* = \frac{d}{c}$$

$$\left| \left(\frac{d}{c} \right)^2 - 25 \right| = 16 \Rightarrow \frac{d}{c} = \pm 3$$

$$1) w = A \frac{z-3}{z+3} \Rightarrow w(9) = A \frac{1}{2} \Rightarrow |A| = \sqrt{\frac{2R}{2}}$$

$$w(0) = -A \Rightarrow |A| = \sqrt{\frac{R}{1}}$$

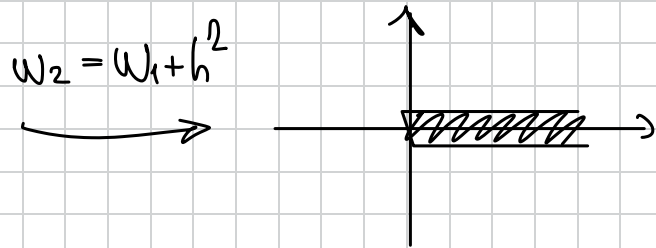
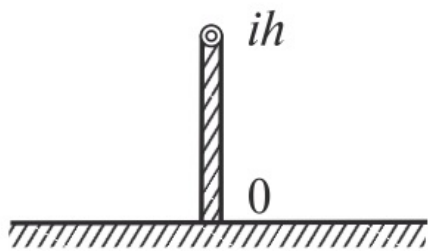
$$2) w = A \frac{z+3}{z-3} \Rightarrow |A| = \sqrt{\frac{R/2}{1/2}}$$

$$|A| = \sqrt{\frac{R}{1}}$$

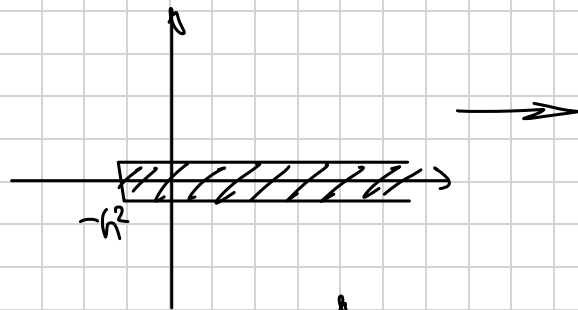
Согласно. пересчит. $R=2 \Rightarrow$

$$\Rightarrow w = 2e^{i\alpha} \frac{z+3}{z-3}$$

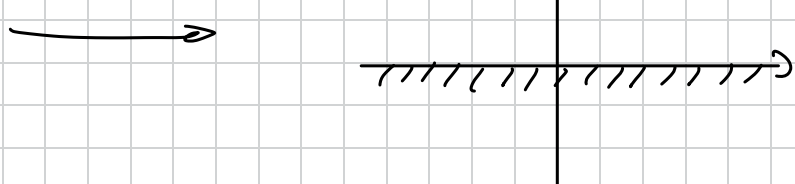
5) 31



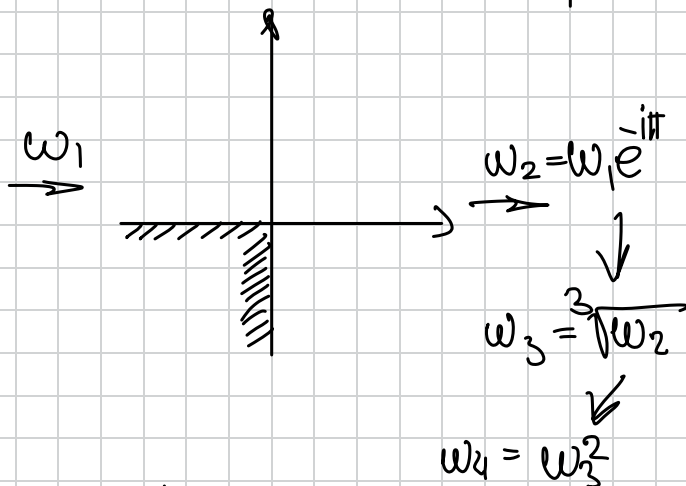
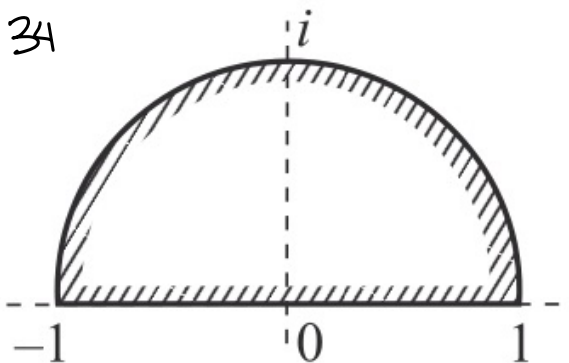
$$w_1 = z^2$$



$$w_3 = \sqrt{|w_2|} e^{\frac{i}{2} \arg w_2}$$



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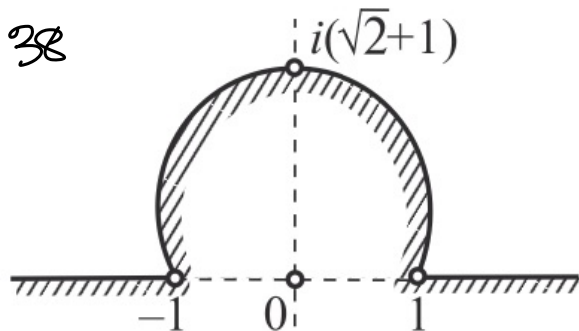
Начиная $w_1(-1) = 0$

$$w_1(1) = \infty$$

$$w_1(0) = -1$$

$$\Rightarrow w_1 = \frac{z+1}{z-1}$$

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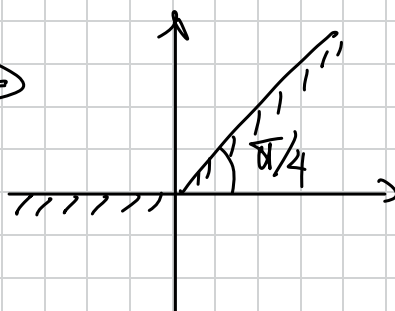


Распространение определено

$$1 \rightarrow 0, -1 \rightarrow \infty$$

$$w_1 = A \frac{z-1}{z+1} \quad \text{Если } \infty \rightarrow 1,$$

$$\text{то } A=1 \Rightarrow w_1 = \frac{z-1}{z+1}$$

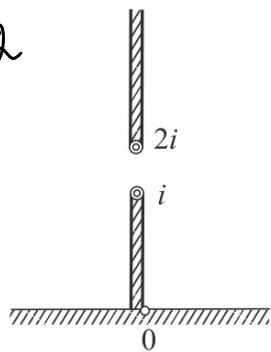


$$w_2 = w_1^{4/3}$$

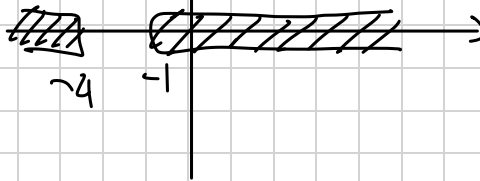
$$w_4 = w_3 e^{-\pi/3}$$

$$w_1([-1, 1]) = (-\infty, 0] \Rightarrow$$

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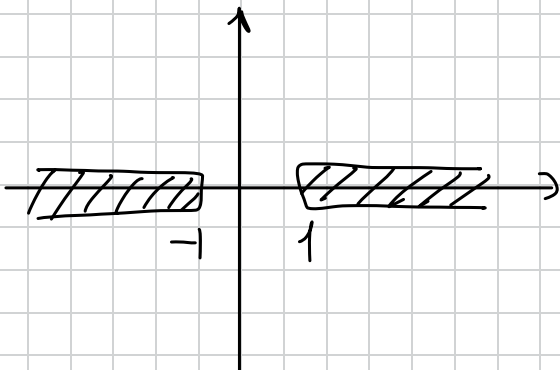
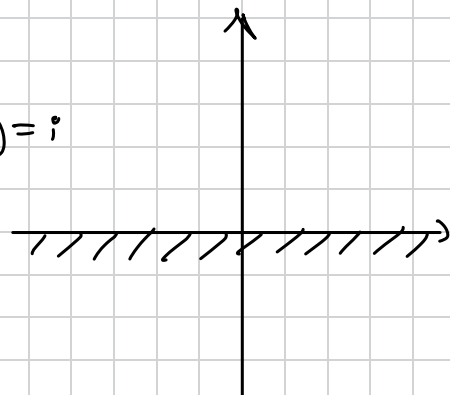
$$\omega_1 = z^2$$



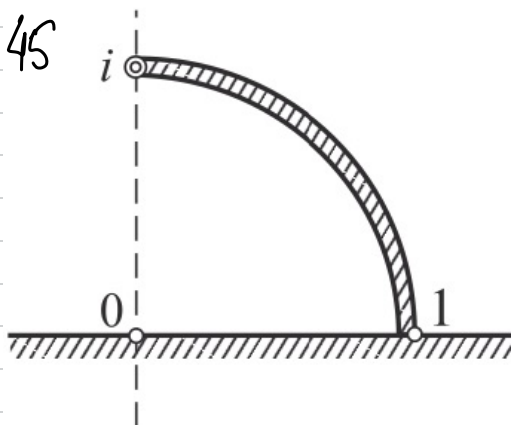
$$\omega_2 = \frac{2}{3}\omega_1 + \frac{5}{3}$$

$$g(\omega_2) \in \sqrt{\omega_2^2 - 1}, \quad g(0) = i$$

$$\omega_3 = \omega_2 + g(\omega_2)$$



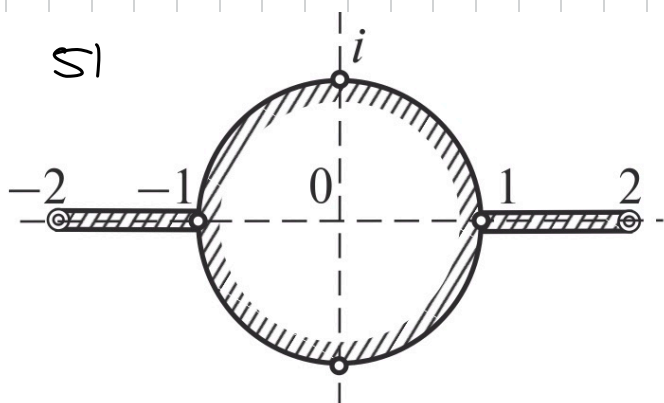
45



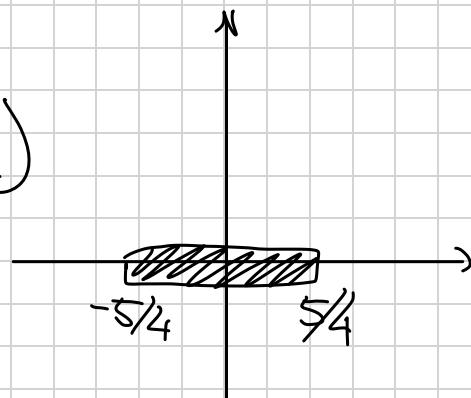
Распространяем отр.: $i \rightarrow i$ $1 \rightarrow 0$
 $-1 \rightarrow \infty$ при продолж. отрезка

$\omega_1 = \frac{z-1}{z+1}$. Далее как со стандартным в 5.31.

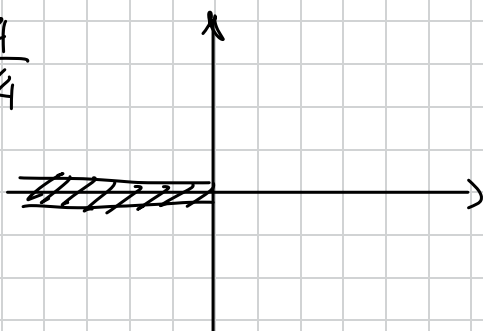
(10)



$$\omega_1 = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

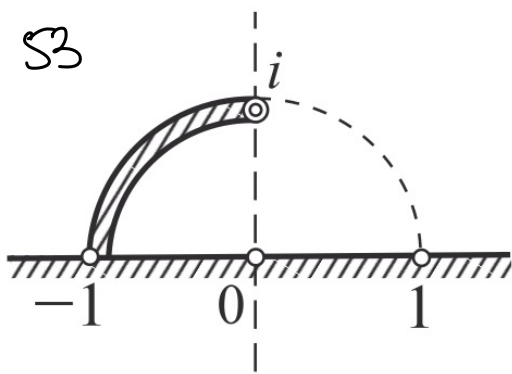


$$\omega_2 = \frac{\omega_1 + 5/4}{\omega_1 - 5/4}$$



$$\omega_3 = -\omega_2 \rightarrow \begin{cases} \omega_4 = \sqrt{\omega_3} \\ \sqrt{-1} = i \end{cases}$$

S3

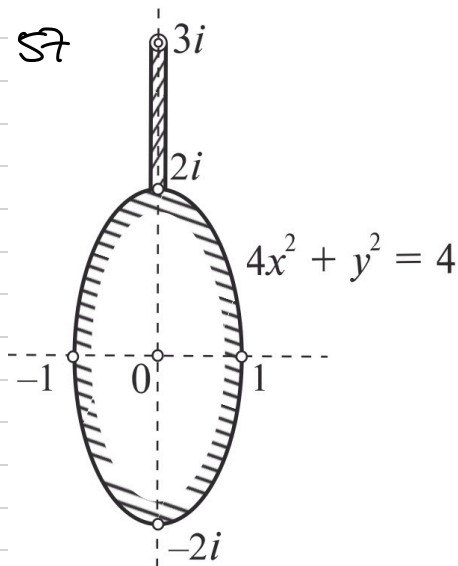


Перевести в стандарт:

$$i \rightarrow i, -1 \rightarrow 0, 1 \rightarrow \infty$$

$$w_1 = \frac{z+1}{-z+1}, \text{ где } z \text{ как стандарт}$$

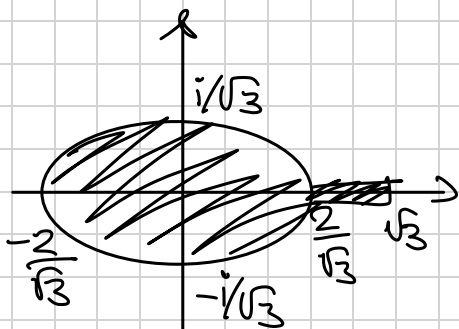
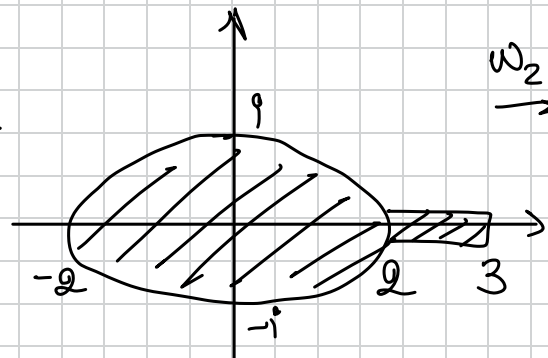
S7



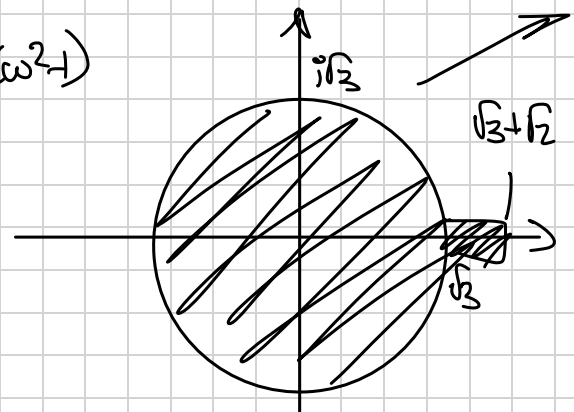
Значит $\Rightarrow$ др. выпр., но черта
границы равна ± 1

$$w_1 = -i\bar{z}$$

$$w_2 = \frac{w_1}{\sqrt{3}}$$



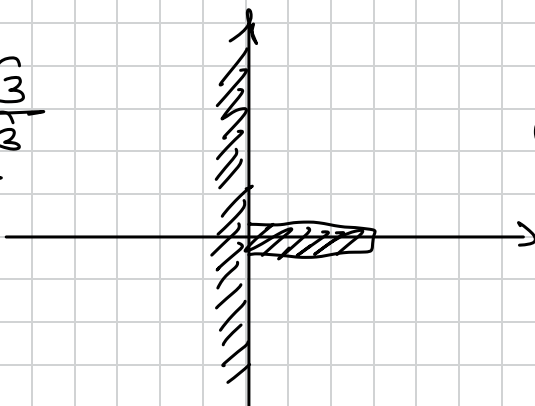
$$w_3 = w_2 + \sqrt{w_2^2 - 1} e^{\frac{i}{2} \arg(w_2^2 - 1)}$$



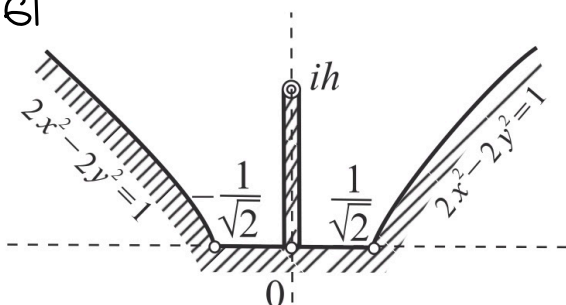
$$w_4 = \frac{w_3 - \sqrt{3}}{w_3 + \sqrt{3}}$$

$$w_5 = iw_4$$

стандарт

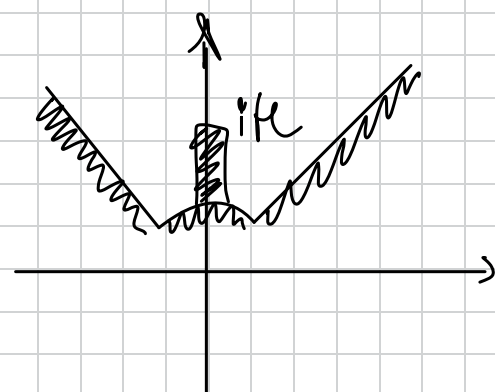


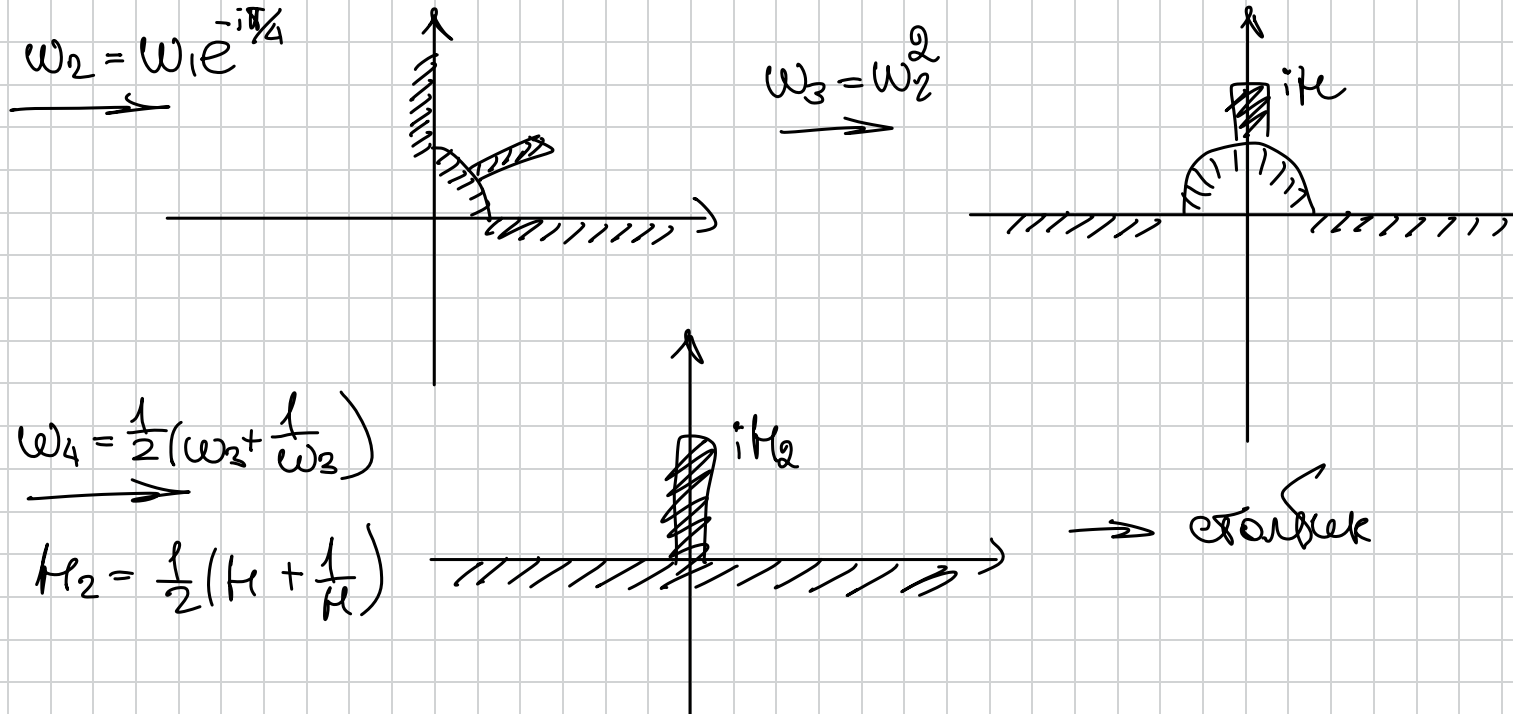
G1



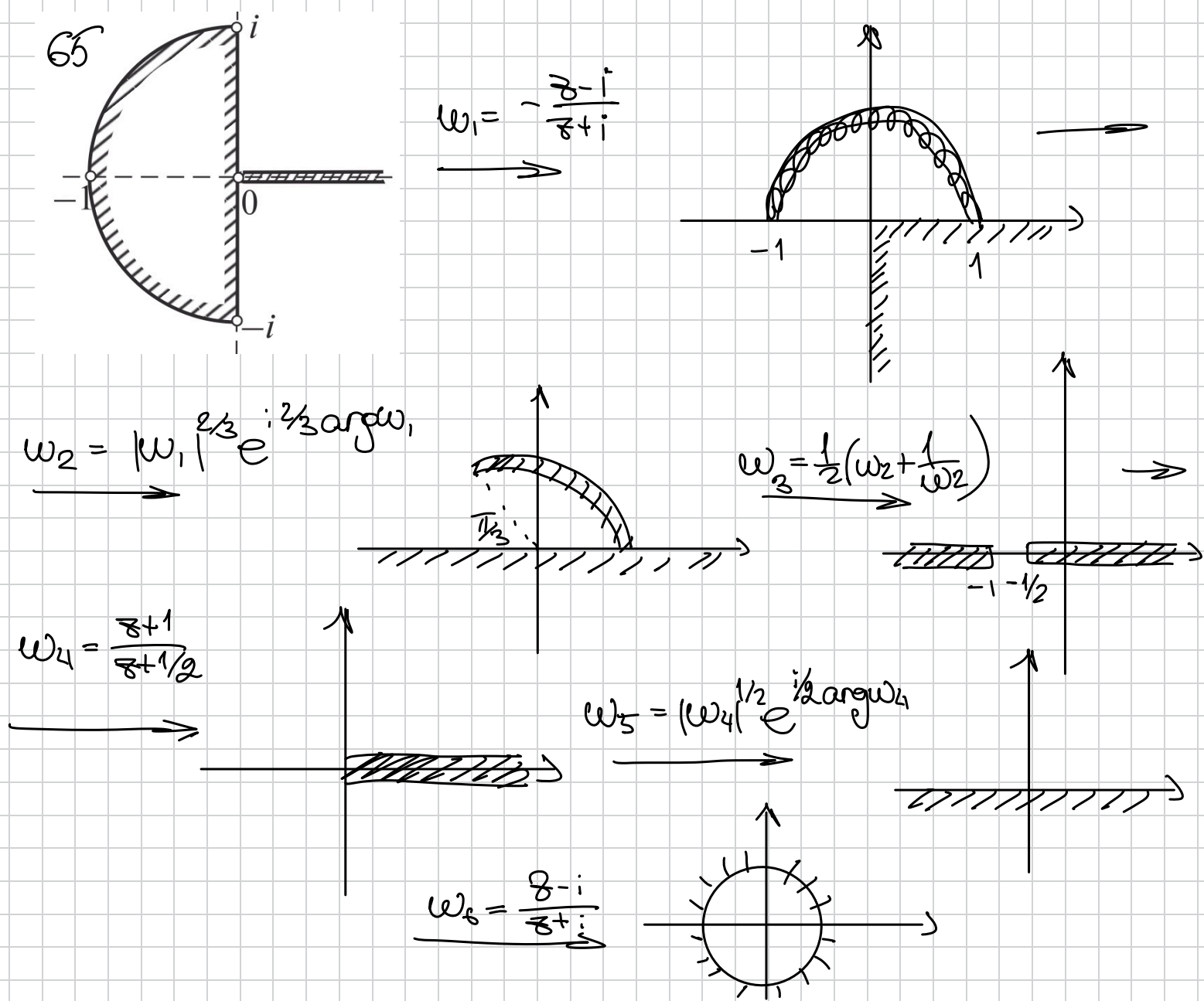
$$w_1 = z + z^2$$

$$ih = i\bar{h} + \sqrt{h^2 - 1}$$

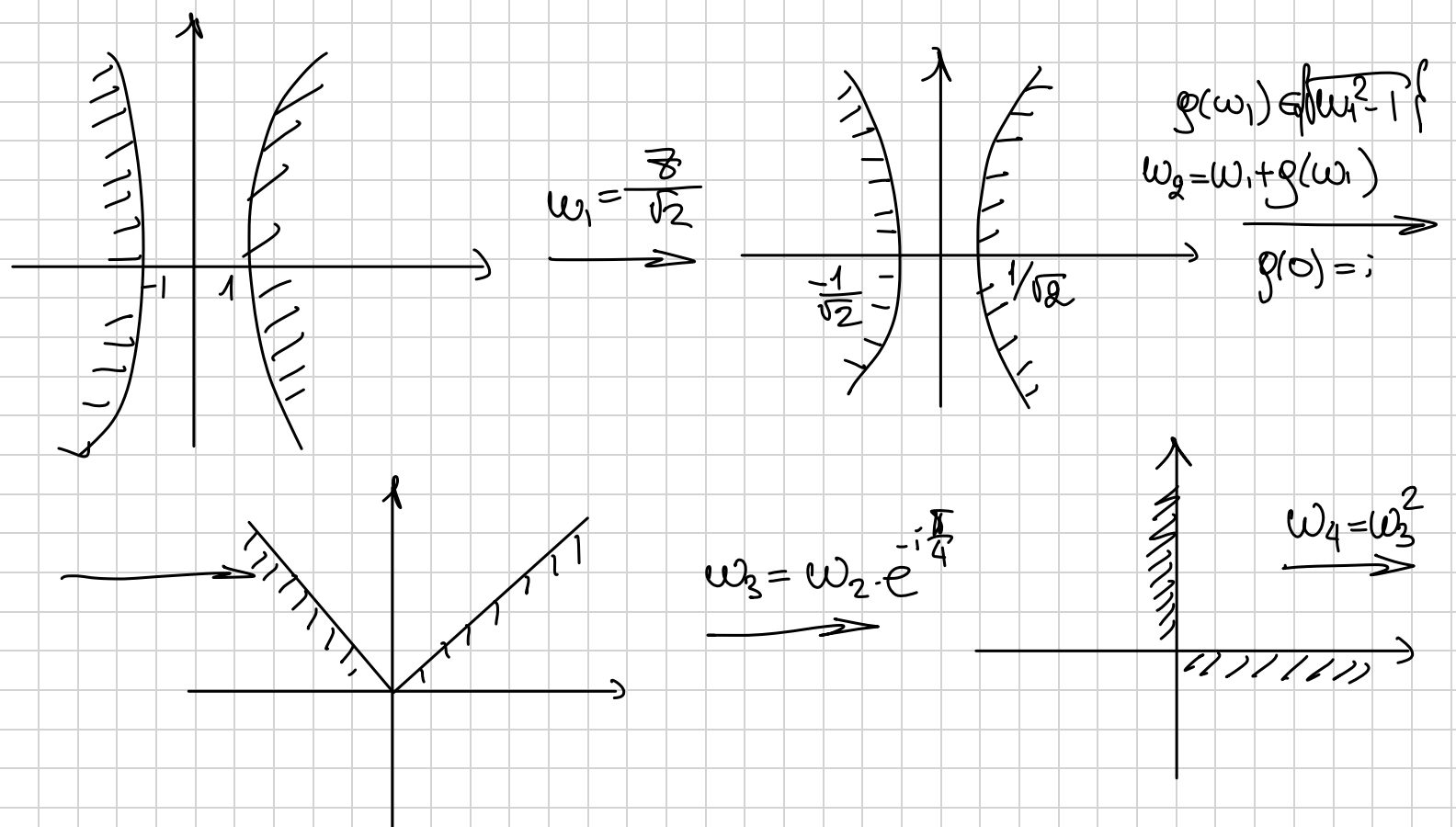




11. Найти какие-либо функции $w(z)$, осуществляющие конформные отображения областей, изображенных на рис. 28.62–28.69, на круг $\{w: |w| < 1\}$.



13. Найти функцию $w(z)$, конформно отображающую область $\{z = x + iy: x^2 - y^2 < 1\}$ на круг $\{w: |w| < 1\}$ и удовлетворяющую условиям $w(0) = 0$, $w(1) = 1$.



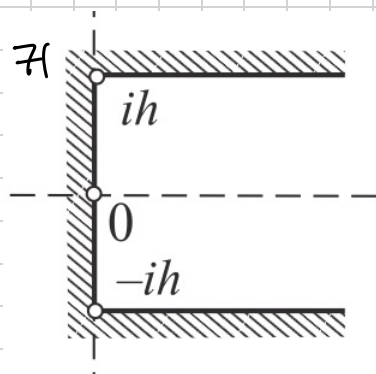
$$w_5 = \frac{w_4 - z_0}{w_4 - \bar{z}_0} e^{i\alpha}; \quad z = 0 \rightarrow 0 \rightarrow i \rightarrow e^{i\pi/4} \rightarrow i \Rightarrow z_0 = i \Rightarrow$$

$$\Rightarrow w_5 = \frac{w_4 - i}{w_4 + i} e^{i\alpha}$$

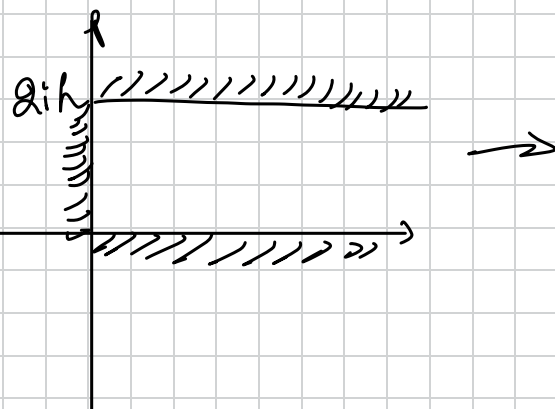
$$z = 1 \rightarrow \frac{1}{\sqrt{2}} \rightarrow e^{i\pi/4} \rightarrow 1 \rightarrow 1 \Rightarrow \frac{1-i}{1+i} e^{i\alpha} = 1 \Rightarrow \alpha = \frac{\pi}{2} \Rightarrow$$

$$\Rightarrow w_5 = i \frac{w_4 - i}{w_4 + i} \Rightarrow \text{перевод в круг}$$

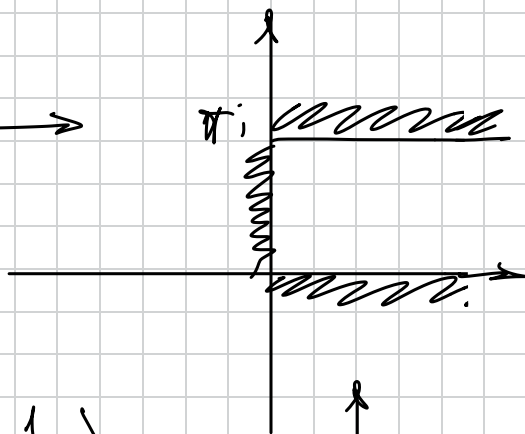
(19)



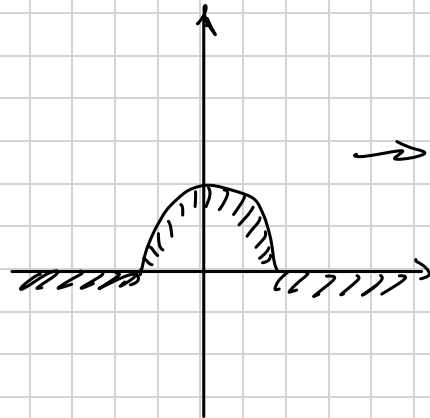
$$w_1 = z + ih$$



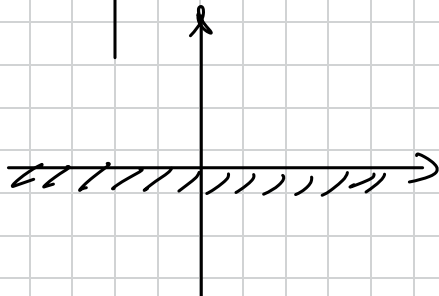
$$\omega_2 = \frac{\pi \omega_1}{2h} \rightarrow$$



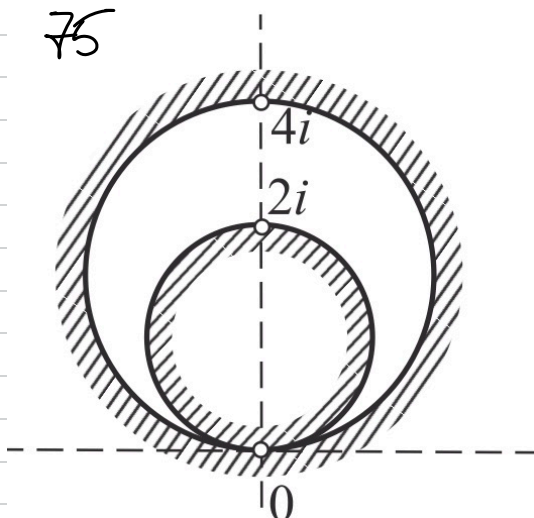
$$\omega_3 = e^{\omega_2} \rightarrow$$



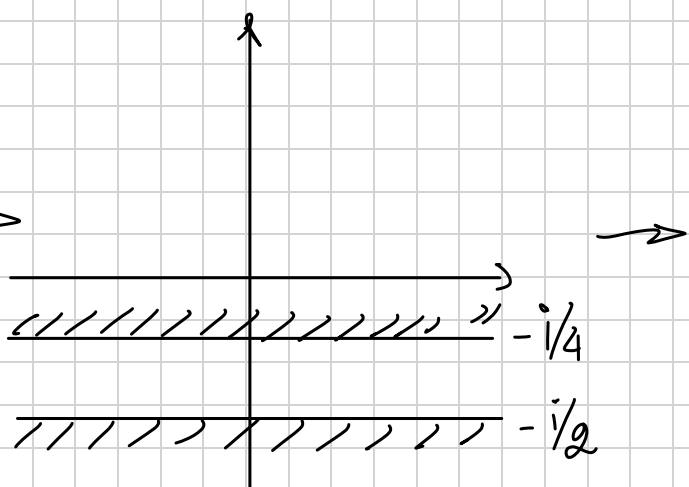
$$\omega_4 = \frac{1}{2}(\omega_3 + \frac{1}{\omega_3}) \rightarrow$$



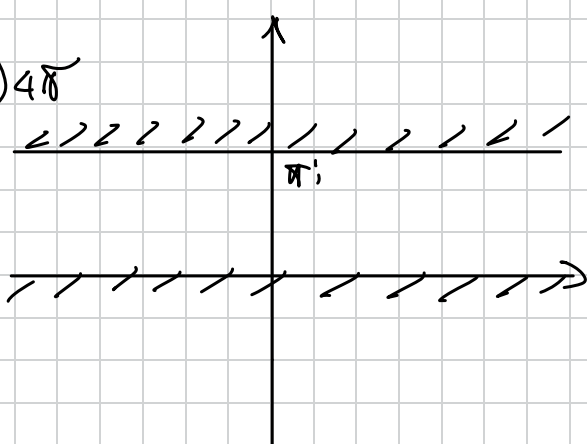
75



$$\omega_1 = \frac{1}{z} \rightarrow$$



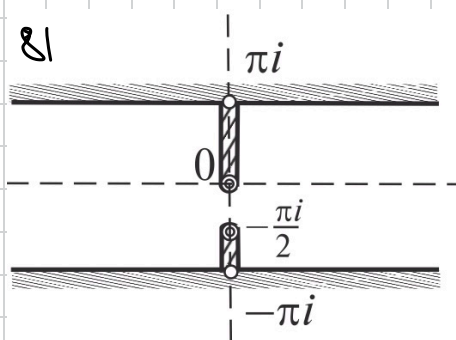
$$\omega_2 = (\omega_1 + \frac{1}{\omega_1}) 4\pi \rightarrow$$



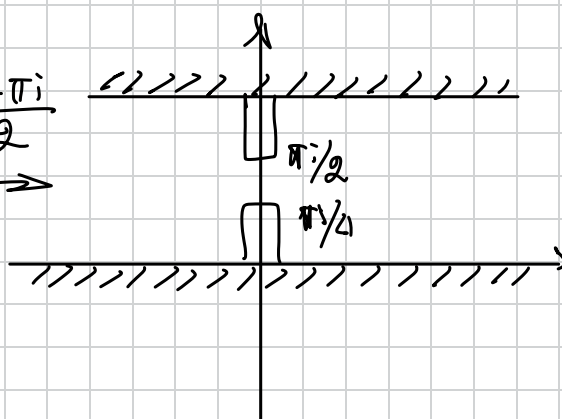
$$\omega_3 = e^{\omega_2} \rightarrow$$



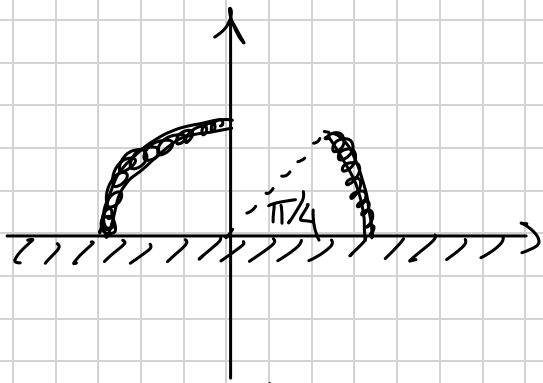
81



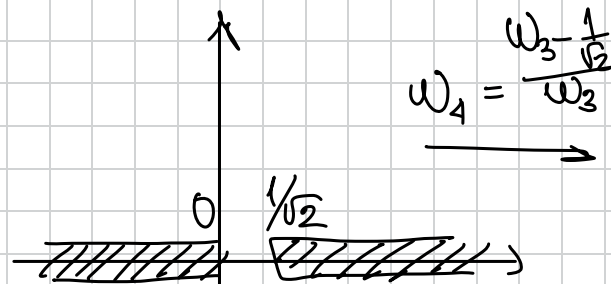
$$\omega_1 = \frac{z + \pi i}{2} \rightarrow$$



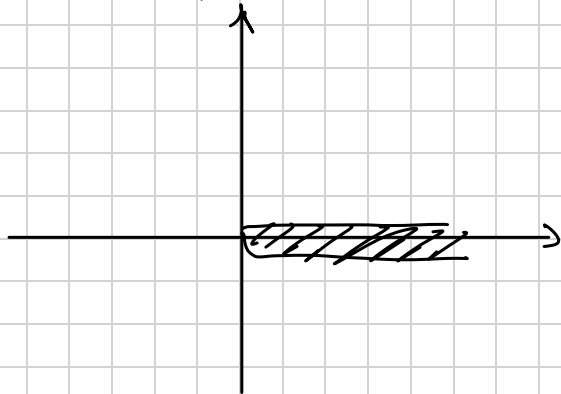
$$\omega_2 = e^{\omega_1} \rightarrow$$



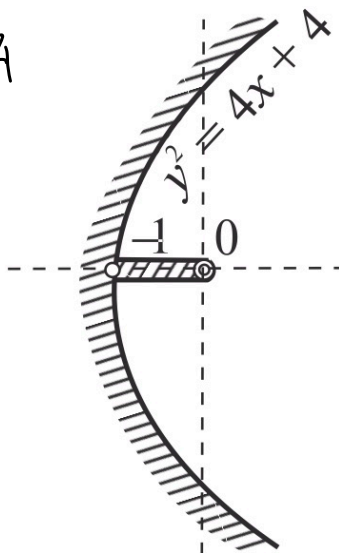
$$w_3 = \frac{1}{2}(w_2 + \frac{1}{w_2})$$



$$w_5 = \sqrt{|w_4|} e^{i/2 \arg w_4}$$

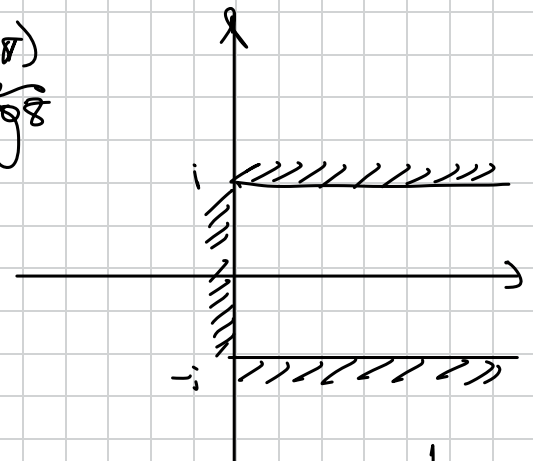


84



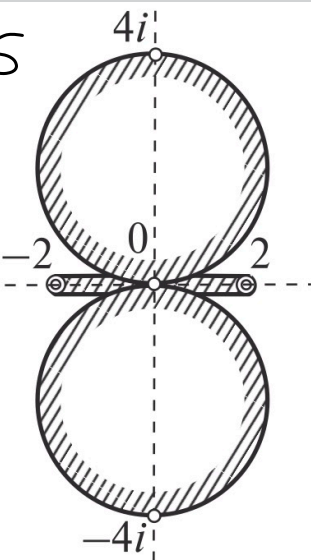
$$w_1 = |z|^{1/2} e^{i/2 \arg z} \quad (-\pi, \pi)$$

$$\sqrt{r} = 1$$

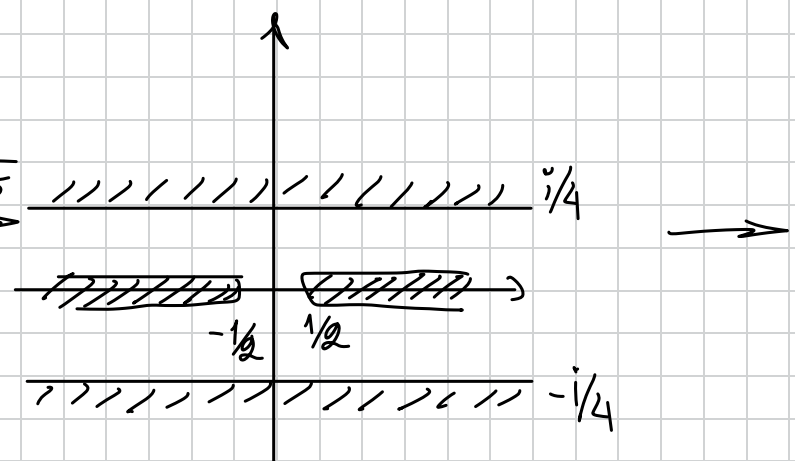


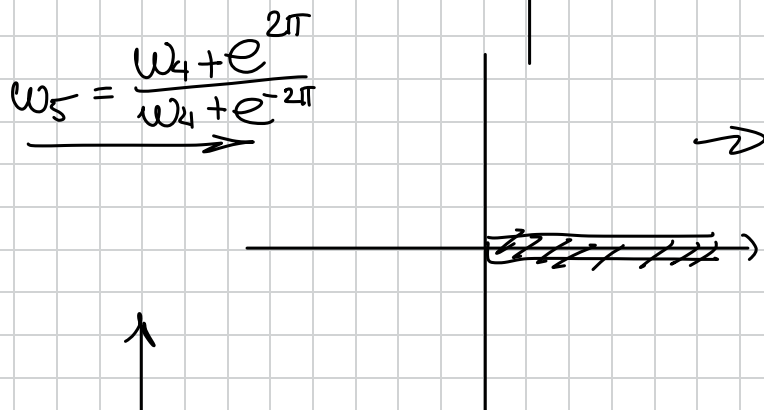
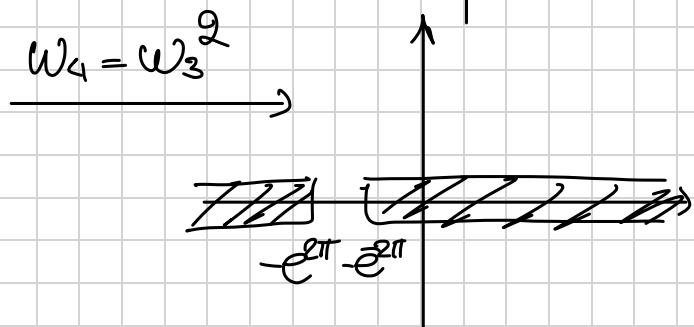
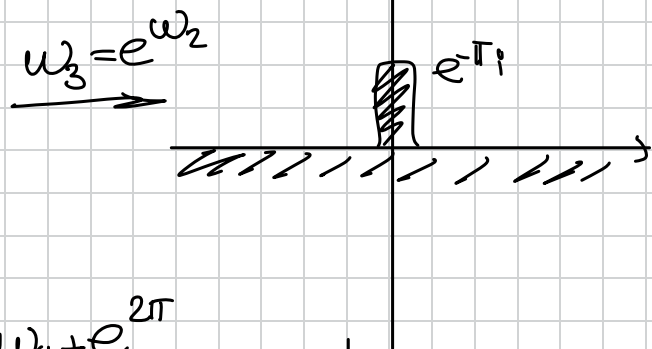
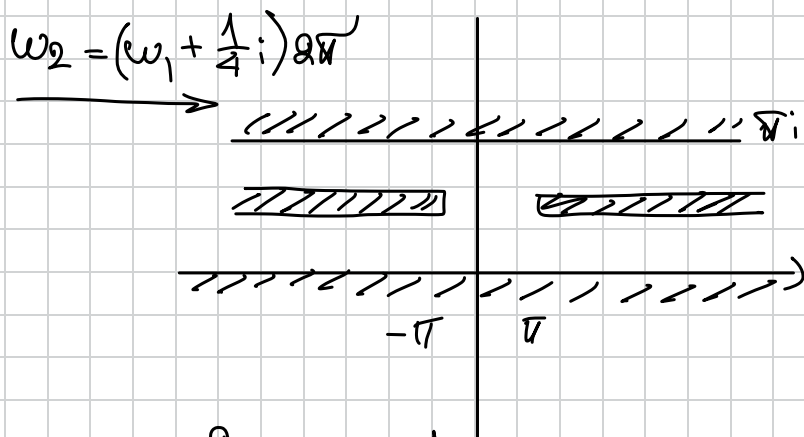
$$w_4 = \frac{1}{2}(w_3 + \frac{1}{w_3}) \leftarrow w_3 = e^{w_2} \leftarrow w_2 = \frac{\pi}{2}(w_1 + i)$$

85

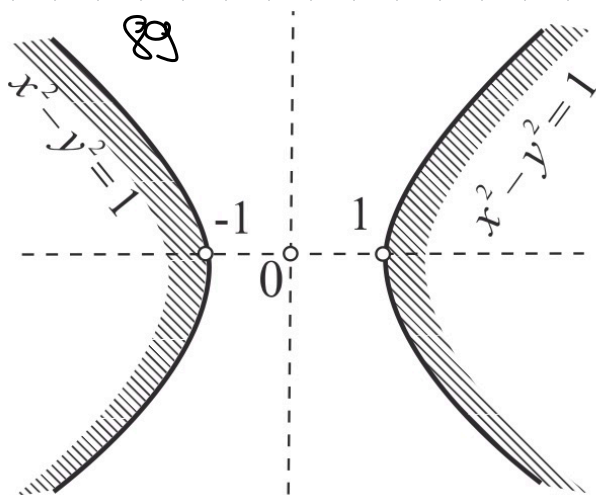
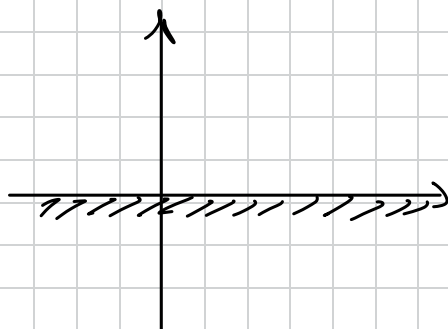


$$w_1 = \frac{1}{8}$$

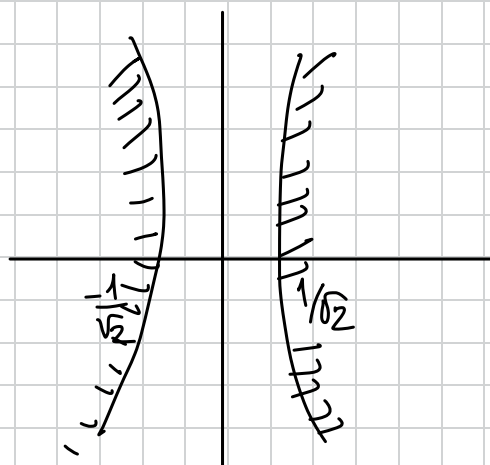




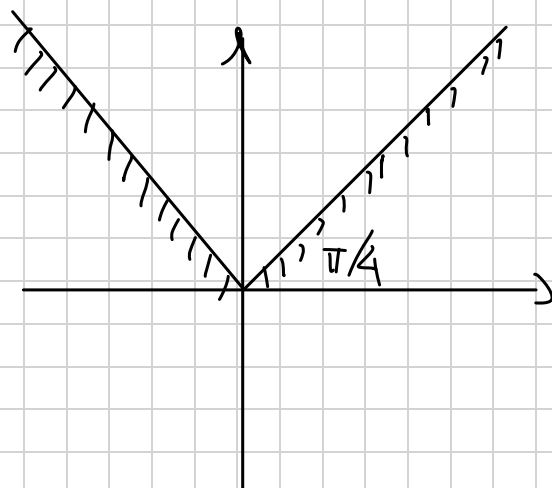
$w_6 = \sqrt{|w_5|} e^{i/2 \arg w_5}, \sqrt{-1} = i$



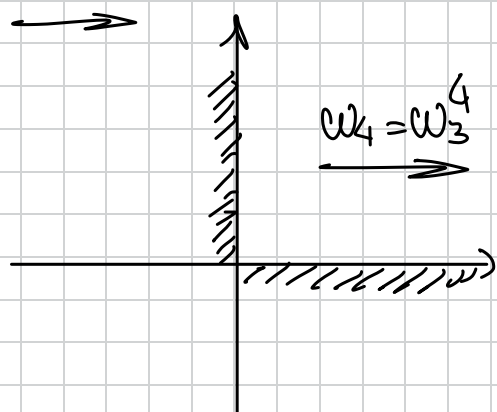
$w_1 = \frac{z}{\sqrt{a}}$

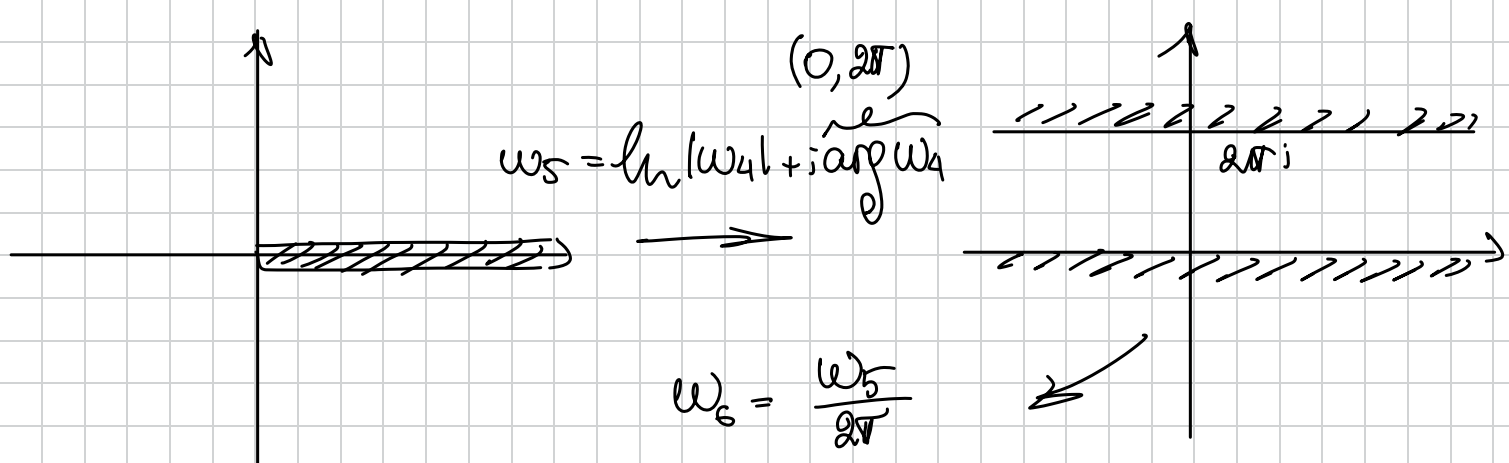


$w_2 = w_1 + g(w_1)$
 $g(w_1) \in \{ \sqrt{w_1^2 - 1} \}$
 $g(0) = i$

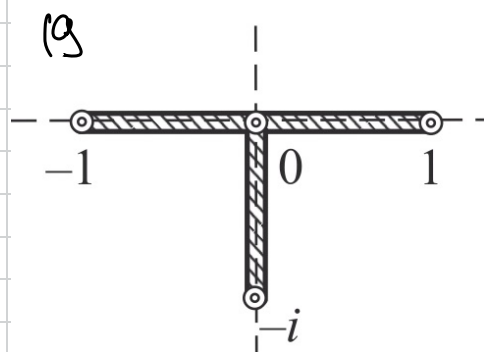


$w_3 = w_2 e^{-i\pi/4}$

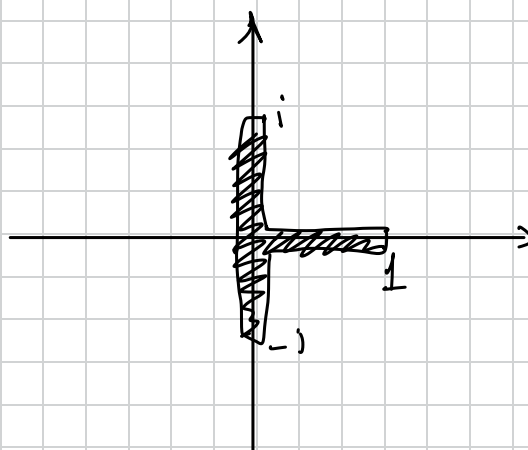




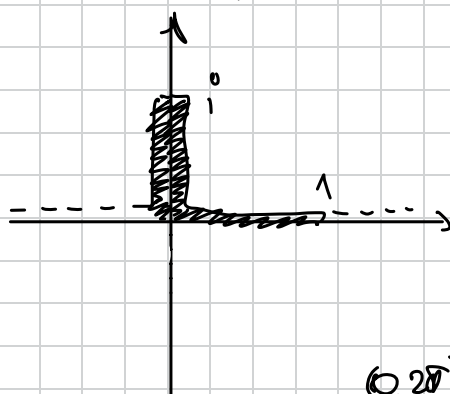
29.3.



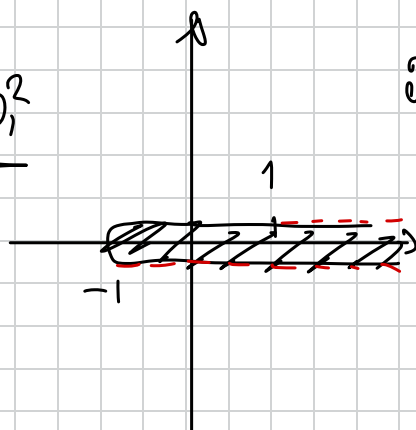
$w_1 = ze^{i\frac{\pi}{2}}$



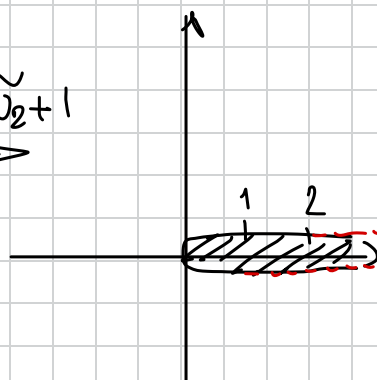
Далее применяем преобразование симм.



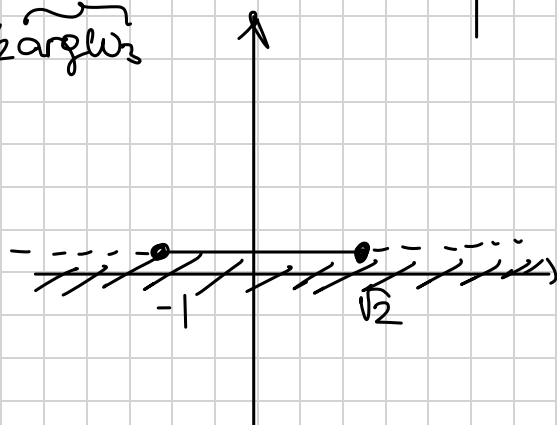
$\tilde{w}_2 = w_1^2$



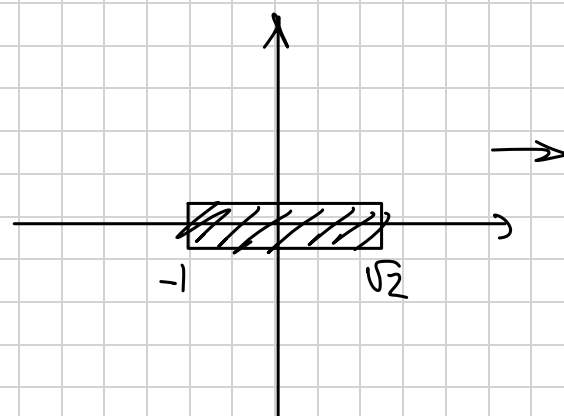
$\tilde{w}_3 = \tilde{w}_2 + 1$



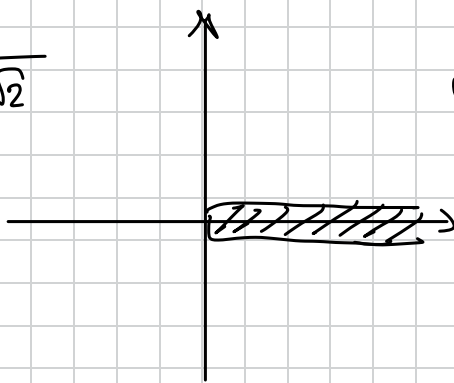
$\tilde{w}_4 = (\tilde{w}_3)^{1/2} e^{i/2 \arg \tilde{w}_3} \quad (0, 2\pi)$



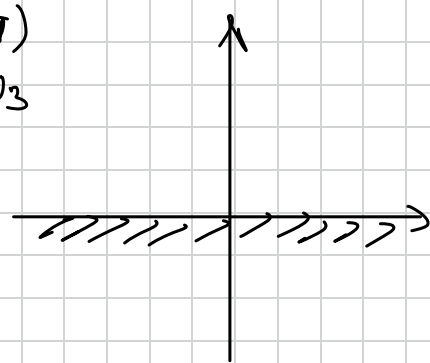
$w_2 = w_4 \circ w_3 \circ w_2 \circ w_1 \rightarrow (\bar{z}), \bar{z} \in \sqrt{\dots}$



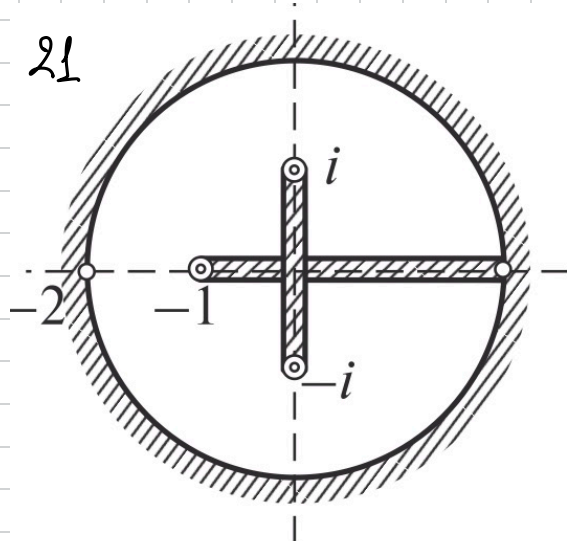
$$\omega_3 = \frac{\omega_2 + 1}{-\omega_2 + \sqrt{2}}$$



$$\omega_4 = |\omega_3|^{1/2} e^{i/2 \arg \omega_3} \quad (0, 2\pi)$$

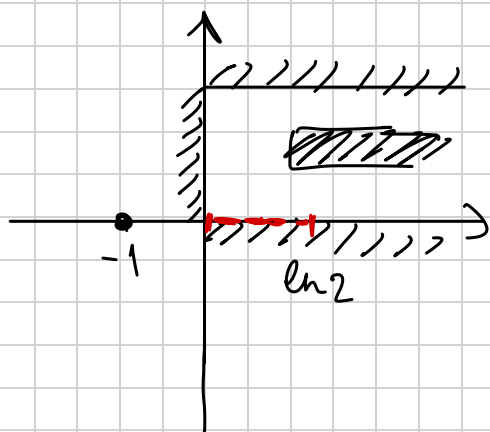
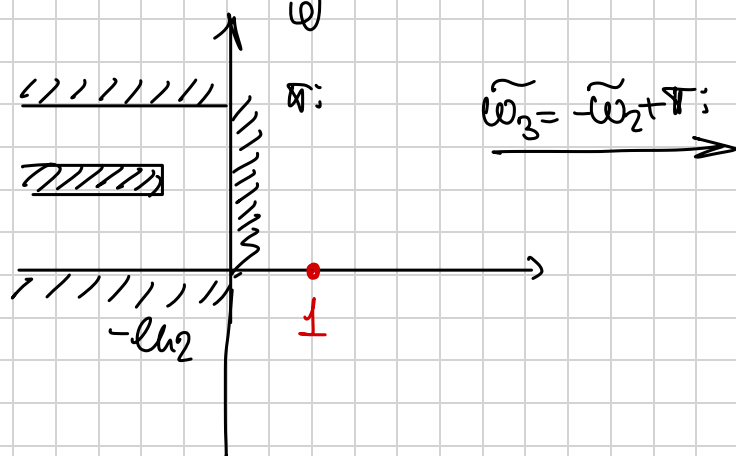


21

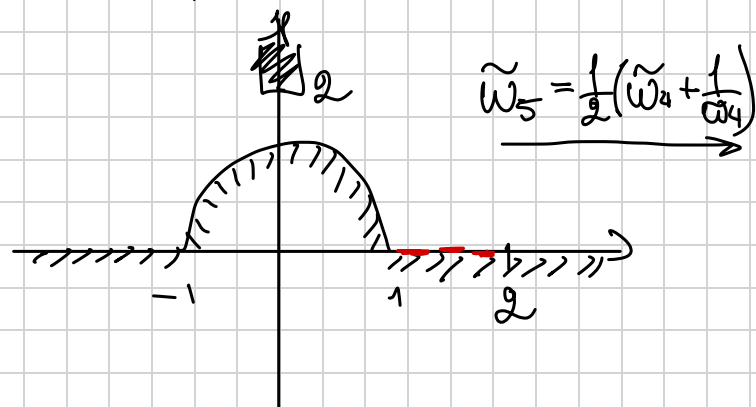


$$\omega_1 = \frac{7}{2}, \text{ given n/purely. complex.}$$

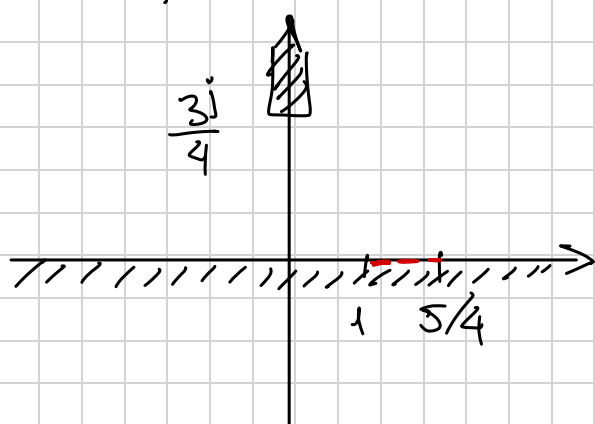
$$\tilde{\omega}_2 = \ln |\omega_1| + i \arg \omega_1$$



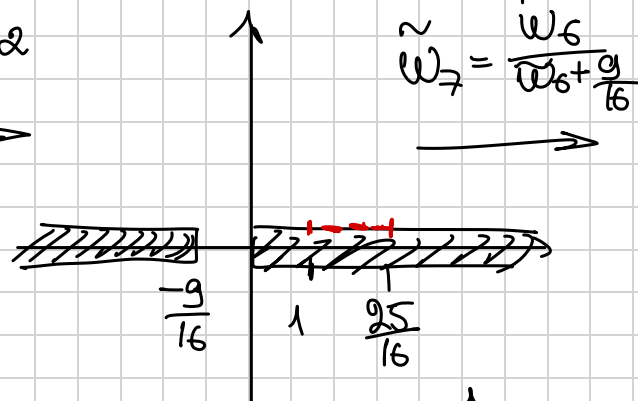
$$\tilde{\omega}_4 = e^{\tilde{\omega}_3}$$



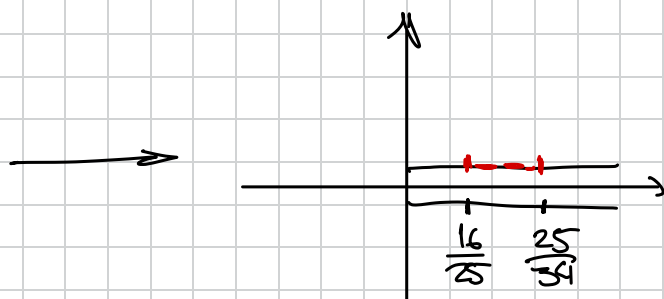
$$\tilde{\omega}_5 = \frac{1}{2}(\tilde{\omega}_4 + \frac{1}{\tilde{\omega}_4})$$



$$\tilde{\omega}_6 = \tilde{\omega}_5^2$$



$$\tilde{\omega}_7 = \frac{\tilde{\omega}_6}{\tilde{\omega}_6 + \frac{9}{16}}$$

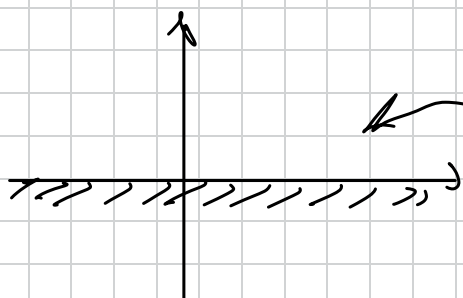
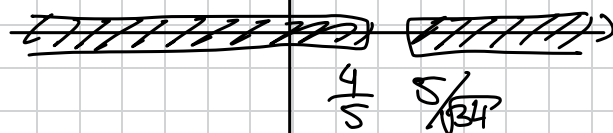


$$\tilde{\omega}_8 = |\tilde{\omega}_7|^{1/2} e^{i/2 \arg \tilde{\omega}_7} \quad (0, 2\pi)$$



Тогда $w_2 = \dots \rightarrow$

$$w_3 = \frac{w_2 - 4/5}{w_2 - 5/\sqrt{34}}$$

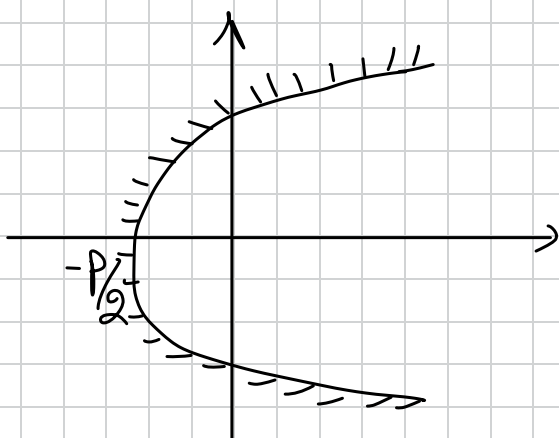


$$w_4 = w_3^{1/2} e^{i/2 \arg w_3}$$

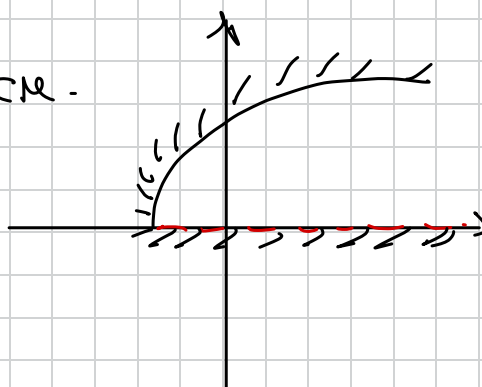
5. Найти какую-либо функцию $w(z)$, конформно отображающую область

$$\left\{ z = x + iy : y^2 < 2p \left(x + \frac{p}{2} \right) \right\}, \quad p > 0,$$

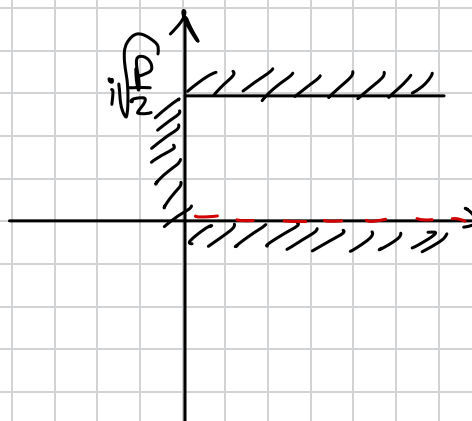
на полуплоскость $\{w : \operatorname{Im} w > 0\}$.



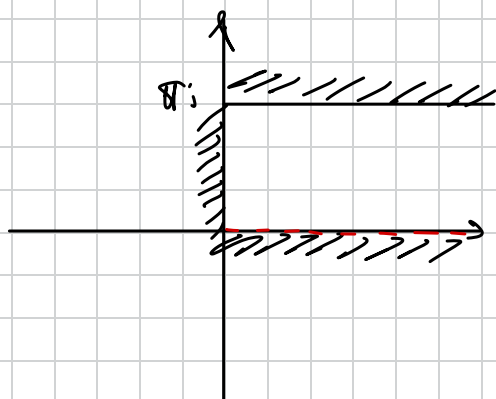
Рассм.



$$\tilde{w}_1 = |z|^{1/2} e^{i/2 \arg z}$$



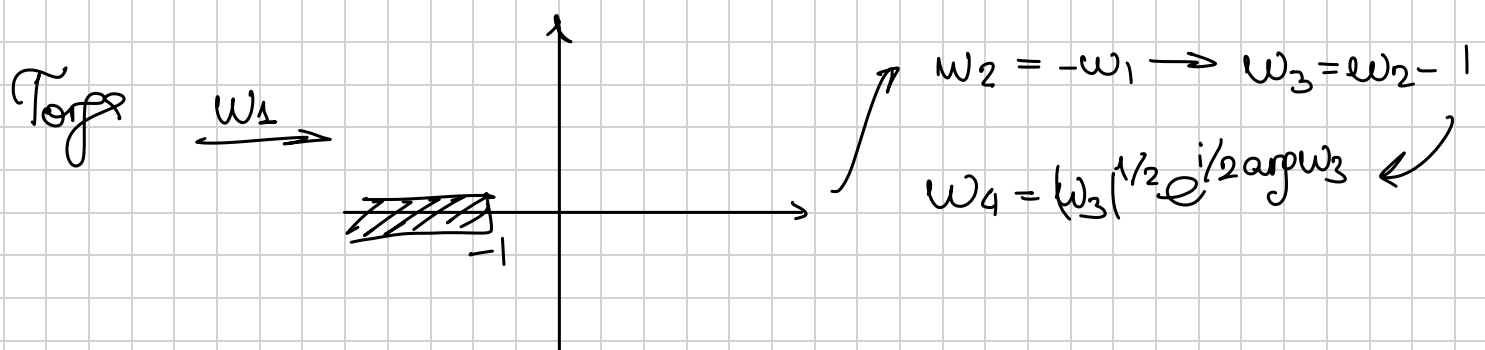
$$\tilde{w}_2 = \left(\frac{2}{p} \right)^{1/2} \tilde{w}_1$$



$$\tilde{w}_3 = e^{\tilde{w}_2}$$

$$w_4 = \frac{1}{2} \left(\tilde{w}_3 + \frac{1}{\tilde{w}_3} \right)$$





$\text{II} \quad a) \int \Delta u = 0, |z| < 1$
 $u|_{|z|=1} = \frac{\sin \theta}{5 + 4 \cos \theta}$

$\text{Nur } \alpha = e^{i\theta} \Rightarrow \sin \theta = \frac{1}{2i} \left(\alpha - \frac{1}{\alpha} \right) \quad \cos \theta = \frac{1}{2} \left(\alpha + \frac{1}{\alpha} \right)$

$$I(z) = \frac{1}{2\pi i} \oint_{|\alpha|=1} \frac{\frac{1}{2i} \left(\alpha - \frac{1}{\alpha} \right)}{5 + 2 \left(\alpha + \frac{1}{\alpha} \right)} \cdot \frac{\alpha + z}{(\alpha - z)\alpha} d\alpha = -\frac{1}{4\pi} \oint_{|\alpha|=1} \frac{(\alpha^2 - 1)(\alpha + z)}{(2\alpha + 1)(\alpha + 2)(\alpha - z)\alpha} d\alpha$$

$$I(z) = \frac{i}{2} \left(\text{res } f(\alpha)_{\alpha=-2} + \text{res } f(\alpha)_{\alpha=\infty} \right)$$

$$\text{res } f(\alpha)_{\alpha=-2} = \frac{3(z-2)}{-3(z+2) \cdot 2} = \frac{2-z}{2(z+2)}$$

$$\text{res } f(\alpha)_{\alpha=\infty} = \lim_{\alpha \rightarrow \infty} \alpha (f(\infty) - f(\alpha)) = -\frac{1}{2}$$

$$I(z) = \frac{i}{4} \frac{-2z}{z+2} = -\frac{i}{2} \frac{z}{z+2}$$

$$u(z) = \text{Re} \left(-\frac{i}{2} \frac{z}{z+2} \right) = \text{Re} \left(-\frac{i}{2} \frac{(x+iy)(x+2-iy)}{(x+2)^2 + y^2} \right) = \frac{y}{(x+2)^2 + y^2} \quad \text{Other: } \leftarrow$$

$$b) \int \Delta u = 0, |z| < 1$$

 $u|_{|z|=1} = \frac{4 + 5 \cos \theta}{(5 + 4 \cos \theta)^2}$

$$I(z) = \frac{1}{2\pi i} \oint_{|\alpha|=1} \frac{4 + \frac{5}{2} \left(\alpha + \frac{1}{\alpha} \right)}{\left(5 + 2 \left(\alpha + \frac{1}{\alpha} \right) \right)^2} \cdot \frac{(\alpha + z)}{(\alpha - z)\alpha} d\alpha =$$

$$= \frac{1}{2\pi i} \oint_{|z|=1} \frac{\frac{1}{2z}(5z^2+8z+5)}{\frac{1}{z^2}(2z^2+5z+2)^2} \cdot \frac{z+z}{(z-z)z} dz = \frac{1}{4\pi i} \oint_{|z|=1} \frac{(5z^2+8z+5)(z+z)}{(z+1)^2(z+2)^2(z-z)} dz =$$

$$= -\frac{1}{2}(\operatorname{res} f(z)_{z=-2} + \operatorname{res} f(z)_{z=\infty})$$

$$\operatorname{res} f(z)_{z=\infty} = \lim_{z \rightarrow \infty} z(f(\infty) - f(z)) = 0$$

$$\operatorname{res} f(z)_{z=-2} = \frac{-((10z+8)(z-2)+9)(z+2) - (z-2)(9-4(2z+1)(z+2))}{9(z+2)^2} =$$

$$= \frac{3(z+2+z-2)}{3(z+2)^2} = -\frac{2z}{(z+2)^2} \Rightarrow I(z) = \frac{z}{(z+2)^2}$$

$$u(z) = \operatorname{Re} \left(\frac{z}{(z+2)^2} \right) = \operatorname{Re} \frac{z(\overline{z+2})^2}{(z+2)^2(\overline{z+2})^2} = \frac{1}{|z+2|^4} \operatorname{Re}(|z|^2 \overline{z} + 4|z|^2 + 4\overline{z})$$

$$= \frac{|z|^2 x + 4|z|^2 + 4x}{(|z+2|^4)} = \frac{(x^2+y^2)(x+4) + 4x}{((x+2)^2 + y^2)^2} \quad \leftarrow \text{erster!}$$